

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2022-2023
Comprehensive Exam (EC-3 Regular)

Course No. : AIMLCZG512

Course Title: Deep Reinforcement Learning

Nature of Exam: Open Book

Weightage: 40%

No. of Pages = 3 ;

No. Of Questions = 6;

Duration : Mins;

Date of Exam: 29-09-2024 (EN)

Note to Students:

1. Answer all the questions.
2. Write your name and sign at the end of all the pages.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1)

[2+2+2 = 6.0 M]

You are designing a reinforcement learning agent for a robot that needs to navigate through a maze to reach the goal. The agent receives a small negative reward (-1) for each step it takes and a large positive reward (+100) for reaching the goal.

- a. Explain how the agent will learn the optimal path through the maze using the concepts of rewards, states, and actions. [2.0 M]
- b. Consider a scenario where the robot has multiple possible paths with varying lengths to reach the goal. How might the agent's discount factor (γ) affect its choice between a longer path with higher immediate rewards and a shorter path with lower immediate rewards? [2.0 M]
- c. If the agent always takes the shortest path, what type of reward structure or learning parameter could you change to encourage exploration of other paths? [2.0 M]

Q2)

[1+2+3+1 = 7.0 M]

A robot is placed in a 3x3 grid world where each grid cell represents a state. The robot can move in four directions (up, down, left, right). It starts in the bottom-left corner (State S1) and needs to reach the top-right corner (State S9). Moving into the walls of the grid keeps the robot in its current state. The robot receives a reward of +10 for reaching State S9, -1 for every step taken, and 0 for all other states. The problem is modeled as a Markov Decision Process (MDP).

s7	s8	s9
s6	s5	s4
s1	s2	s3

The transition probability for each action is as follows:

- 80% chance the robot moves in the intended direction.
- 20% chance the robot moves in a random direction (not the intended one).

a. State Transitions and Rewards:

- (i) Define the state-transition matrix for the robot's movement. [1.0 M]
- (ii) Calculate the expected reward for taking an action from S1 (bottom-left corner) when the robot chooses to move to the right. [2.0 M]

- b. **Value Iteration:** Suppose the discount factor γ is 0.9. Perform one iteration of value iteration for the states S1, S2, and S3. Assume the initial value for each state is 0. [3.0 M]
- c. **Policy Selection:** Based on the above setup, what kind of policy would you suggest for the robot if its goal is to maximize long-term rewards? [1.0 M]

Q3)

[1+2+1.5+ 2.5 = 7M]

- (a) Write down a scenario where you will find Monte Carlo methods more appropriate than dynamic programming-based solutions. [0.5 M] Also, write down a scenario when dynamic programming solutions are preferred over Monte Carlo-based solutions. [0.5 M]
- (b) Write down any two ways (with reasoning) that increase the coverage of states by episodes being used to learn solutions based on Monte Carlo. [2.0 M] [Precise statements]
- (c) If there are k actions ($a_1, a_2, a_3, \dots, a_k$), with one being the greedy (say a_g) what will be the ϵ -greedy policy when $\epsilon = 0.2$. [1.5 M] [General expressions are discouraged. Provide an answer that exactly matches this scenario]
- (d) You have a simple system with **2 states**, **S1** and **S2**. The agent (a robot) starts in **S1** and follows a random policy to transition between the two states. The goal is to estimate the value function **V(S1)** using **First-Visit Monte Carlo**.

States:

- **S1:** Starting state (no reward)
- **S2:** Terminal state (reward = +1)

Transition Rules:

- From **S1**, the agent has a 50% chance of staying in **S1** and a 50% chance of moving to **S2**.
- When the agent reaches **S2**, the episode ends with a reward of +1.

Goal: Estimate the value function **V(S1)** using First-Visit Monte Carlo after 4 episodes. [2.5 Marks]

Q4)

[4+1+1+1 = 7M]

- a. Consider a Gridworld where we want to use **Temporal Difference (TD)** Learning to estimate the values of the states. Suppose we observe the following state-action transitions and rewards - (W, East, X, 3), (X, South, Z, 5), (X, East, V, 1), (W, East, X, 3). Assume the initial values of each state: **V(X)=V(Y)=V(Z)=V(W)=1**, the discount factor ' **$\gamma=0.5$** ' and learning rate ' **$\alpha=0.5$** '. **Hint:** The TD(0) rule: $V(S) = V(S) + \alpha(R_{t+1} + \gamma V(S') - V(S))$
- What are the learned values for each state after applying TD(0) learning for all four observations for the first iteration? [4.0 M]
 - Describe a situation where TD learning might be preferred over Monte Carlo methods, and vice versa. [1.0 M]
- b. Imagine a scenario where you are programming a self-driving car to navigate through a city. The car must learn to make decisions at every intersection, accounting for traffic signals, pedestrians, and other cars. You want to use reinforcement learning to help the car develop an optimal policy.
- In this scenario, what challenges would you face if you used tabular learning to represent every possible state the car might encounter? [1.0 M]
 - How would using a parameterized function approximator help you overcome these challenges, especially when dealing with a large or continuous state space like city navigation? [1.0 M]

Q5)

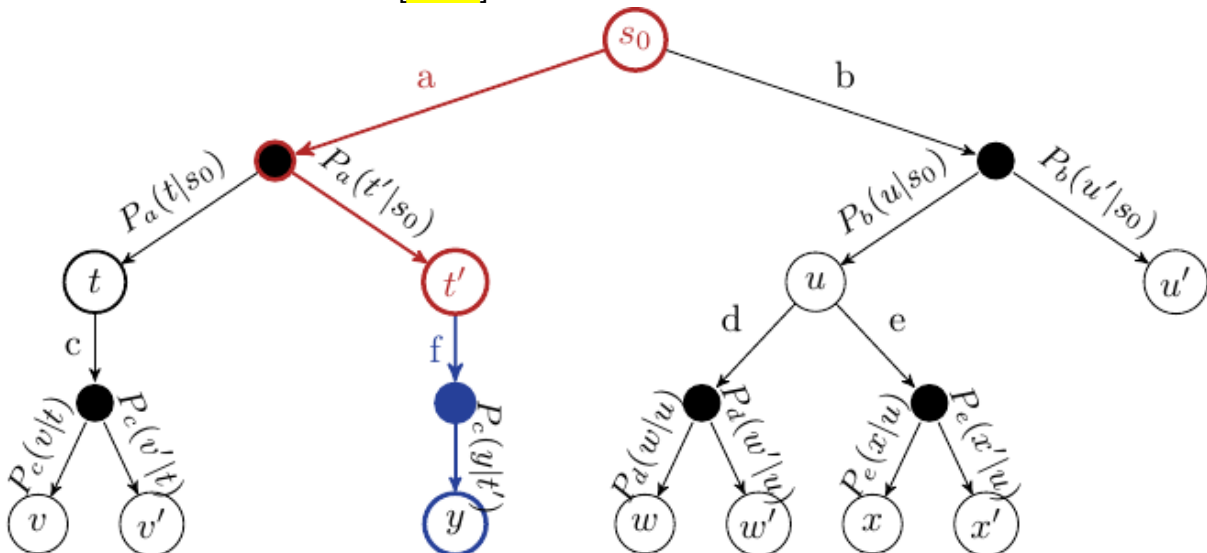
[3+ 2.5 = 5.5 Marks]

- a. You are developing an intelligent agent to play a complex video game where the agent must learn to navigate through various levels, collect items, and defeat enemies. The game environment is dynamic, with many possible states and actions, making the learning process challenging.
- In this scenario, how will the **REINFORCE** algorithm enable the agent to learn an optimal policy for navigating the game? Identify one potential drawback of using REINFORCE alone in such a dynamic environment. [1.0 M]
 - How would implementing an **actor-critic** method improve the agent's learning compared to using **REINFORCE**? State the roles of the actor and the critic, and how their interaction enhances the learning process in the context of the video game environment. [1.0 M]
 - Contrast the learning stability and convergence speed of the two methods in this scenario. How might the variance in updates differ between **REINFORCE** and the **actor-critic** approach, and what implications does this have for the agent's performance? [1.0 M]
- b. How does the REINFORCE algorithm use the policy gradient theorem? [1.0 M] Explain why the REINFORCE algorithm uses the policy gradient theorem in a particular way. [1.5 M]

Q6)

[4.5 + 3 = 7.5 Marks]

- (a) Consider an RL agent needs to make an action selection at s_0 using MCTS. From all the states, the actions are equally likely. Assume $Q(s_0, a)$, $Q(t', f)$ are 12 and 9 respectively. The state y has been evaluated with a rollout policy with a return of 25. What are the updated values backed up for $Q(s_0, a)$, $Q(t', f)$? Assume $\gamma = 0.6$. The notations used are the same as the ones used in the class [4.5 M].



- (b) Consider an agent playing a game of chess. Would you recommend using behavior cloning to learn this agent? [1 M] Why or why not? Provide any 2 reasons with details [2.0 M]